

Lecture 1 — Writing a Linear Program

Course notes: from a problem in words to a mathematical model

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What you will learn. After this lecture you can read a problem described in words and write it as a *linear program*: you choose the unknowns, you write what you want to maximise or minimise, and you write the limits as equations or inequalities. You will also learn to write the same program in different forms. Solving linear programs is the subject of Lecture 2.

How to read these notes. Each idea starts with an example with numbers. Definitions and rules are in boxes. The "Check yourself" questions have their answers at the end of the notes.

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1 What is operations research?

Operations research (OR) uses mathematics to make the *best decision* when resources are *limited*: time, money, material, energy. The work always has two steps:

1. write a *model*: translate the problem into mathematics;
2. *solve* the model, usually with an algorithm.

Linear programming (LP) is the most used model of operations research. In a linear program, all expressions are *linear*: numbers multiplied by unknowns, then added, like $40x_1 + 60x_2$.

A little history: L. Kantorovich wrote the first linear programs in 1939 to plan production; G. Dantzig invented the simplex method, the main algorithm to solve them, in 1947; Kantorovich and T. Koopmans received the Nobel Prize in Economics in 1975 for this work.

Linear programming is used in factories (how much to produce), in industry (how to mix raw materials), in logistics (which warehouse supplies which customer), for planning over time (production, stocks, stored energy) and for schedules. Companies that use robots or drones meet the same questions: which robot does which task, where to send spare parts, when to charge the batteries.

2 A first example, step by step: the drone workshop

2.1 The problem

A small workshop assembles two models of drones: model A (simple) and model B (with a camera). The data are:

	Model A	Model B	Available per week
Assembly (hours for one drone)	1	2	40 hours
Testing (hours for one drone)	3	2	60 hours
Cameras (for one drone)	0	1	18 cameras
Profit (thousand DA for one drone)	40	60	

Question: how many drones of each model should be made each week to get the largest profit?

To write the model we answer three questions: *What do we decide? What do we want? What limits us?*

2.2 Question 1: what do we decide? The variables

We must decide two numbers. We give them names:

- x_1 = number of drones A made per week;
- x_2 = number of drones B made per week.

These unknowns are the *decision variables*. Drones are counted in whole numbers; we explain in Section 2.5 how the model treats this. A choice of values is a *plan*: for example $(x_1, x_2) = (10, 5)$ means "make 10 drones A and 5 drones B each week".

Important habit. Always write in a full sentence what each variable means, with its unit (number of drones, kg, hours, ...) and its period (per day, per week, ...).

2.3 Question 2: what do we want? The objective

One drone A gives 40 thousand DA (Algerian dinars), so x_1 drones A give $40x_1$ thousand DA. One drone B gives 60 thousand DA, so x_2 drones B give $60x_2$ thousand DA. We count the profit in thousands of dinars to keep small numbers. The total profit is

$$z = 40x_1 + 60x_2 \qquad \text{(to maximise)}.$$

Example. For the plan $(10, 5)$: $z = 40 \times 10 + 60 \times 5 = 400 + 300 = 700$ thousand DA, that is 700 000 DA.

Units. (thousand DA per drone) $\times$ (number of drones) = thousand DA.

The function z is the *objective function*. We *maximise* it when it is a profit or a gain; we *minimise* it when it is a cost, a time or an energy.

2.4 Question 3: what limits us? The constraints

- **Assembly.** One drone A needs 1 hour, so x_1 drones A need $1 \times x_1 = x_1$ hours. One drone B needs 2 hours, so x_2 drones B need $2x_2$ hours. In total $x_1 + 2x_2$ hours, and we have at most 40 hours:
$$x_1 + 2x_2 \leq 40.$$
- **Testing.** In the same way: $3x_1 + 2x_2 \leq 60$.
- **Cameras.** Each drone B needs one camera, drones A need none, and we have at most 18 cameras: $x_2 \leq 18$.
- **Signs.** We cannot make a negative number of drones: $x_1 \geq 0$ and $x_2 \geq 0$. These are the *sign constraints*. Never forget them.

2.5 The complete model

$$\begin{aligned} \max \quad & z = 40x_1 + 60x_2 && \text{(profit, thousand DA per week)} \\ \text{subject to} \quad & x_1 + 2x_2 \leq 40 && \text{(assembly hours)} \\ & 3x_1 + 2x_2 \leq 60 && \text{(testing hours)} \\ & x_2 \leq 18 && \text{(cameras)} \\ & x_1 \geq 0,\ x_2 \geq 0 && \text{(signs)} \end{aligned}$$

"Subject to" is often written "s.t.".

Whole numbers or real numbers?

Drones are whole objects. So we must distinguish two problems:

- the *real problem*: the same objective and constraints, with $x_1, x_2 \in \{0, 1, 2, \dots\}$ (whole numbers). A problem with whole-number variables is called an *integer program*;
- its *LP relaxation*: the linear program above, where x_1, x_2 can be any real numbers ≥ 0.

In this course we solve the LP relaxation, because it is much easier. Section 2.7 explains when its answer is also the answer of the real problem.

Definition 1.1 — Linear program

A *linear program* (LP) is a problem where we maximise or minimise a linear function of the variables, subject to linear constraints. Each constraint is a linear expression which is "$\leq$", "$=$" or "$\geq$" a number.

2.6 Vocabulary

Definition 1.2 — Feasible and optimal solutions

- A *feasible solution* is a plan that satisfies *all* the constraints (including the sign constraints).
- The *feasible set* is the set of all feasible solutions.
- An *optimal solution* is a feasible solution with the best value of the objective (the largest for a maximisation, the smallest for a minimisation). This best value is the *optimal value*.

Example 1. Is $(10, 10)$ feasible? Assembly: $10 + 2 \times 10 = 30 \leq 40$, yes. Testing: $3 \times 10 + 2 \times 10 = 50 \leq 60$, yes. Cameras: $10 \leq 18$, yes. Signs: yes. So $(10, 10)$ is feasible, with $z = 40 \times 10 + 60 \times 10 = 1000$ thousand DA (one million DA).

Example 2. Is $(15, 15)$ feasible? Assembly: $15 + 30 = 45 > 40$. No. One violated constraint is enough: we do not need to check the others.

Testing some plans.

Plan (x_1, x_2)	Assembly (≤ 40)	Testing (≤ 60)	Cameras (≤ 18)	Feasible?	z (thousand DA)
(0, 0)	0	0	0	yes	0
(20, 0)	20	60	0	yes	800
(0, 18)	36	36	18	yes	1080
(10, 10)	30	50	10	yes	1000
(15, 15)	45	75	15	no	—
(4, 18)	40	48	18	yes	1240

The best plan in this table gives 1240 thousand DA. But the variables can take infinitely many values, so we cannot test all the plans: we need a method. This is the subject of Lecture 2.

Check yourself

Q1. Is the plan $(12, 12)$ feasible? If yes, what is its profit?
Q2. Is the plan $(20, 2)$ feasible?

A plan is a point of the plane. With two variables, each plan (x_1, x_2) is a point. Each constraint is a line, and the feasible plans form a region (Figure 1). The plan $(15, 15)$ is outside the region, because it violates the assembly constraint. The best plans of the table are on the border of the region. Finding the best plan means finding the best point of the region: this is what the graphical method of Lecture 2 does.

2.7 Whole numbers: lucky or not?

In Lecture 2 we will find that the best plan of the LP relaxation is $(x_1, x_2) = (10, 15)$, with $z = 1300$ thousand DA. These are whole numbers, so this plan is also the best plan of the real problem: no

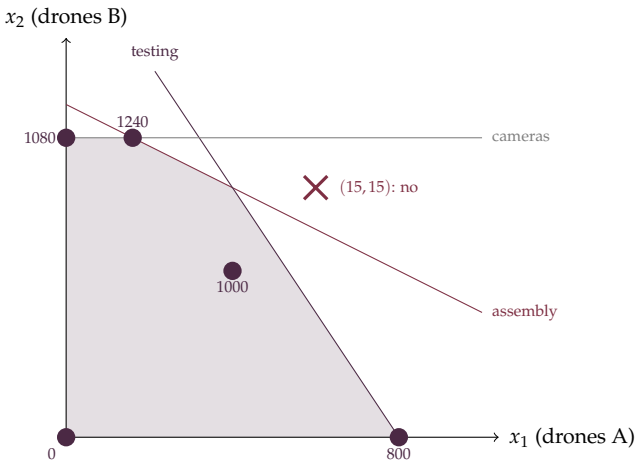


Figure 1: The plans of the table as points, with their profit. The shaded region contains the feasible plans; the cross $(15, 15)$ is not feasible.

whole-number plan can beat the best real-number plan. Here we are *lucky*: this is a coincidence, and it is *not guaranteed* for other production problems.

Example. Suppose the workshop has 61 testing hours instead of 60. Then:

	Plan (x_1, x_2)	z (thousand DA)
Best plan of the LP relaxation	$(10.5,\ 14.75)$: not whole numbers	1305
Rounding up	$(11, 15)$: assembly $11 + 30 = 41 > 40$, not feasible	—
Rounding down	$(10, 14)$	1240
Best whole-number plan	$(10, 15)$	1300

So we cannot simply round the LP answer: rounding can give a plan that is not feasible, or a plan that is not the best. When whole numbers are necessary, we check the LP answer, and if it is not made of whole numbers we need *integer programming* (another subject). The LP relaxation is still very useful: its optimal value (1305) is an upper bound for the real problem, because every whole-number plan is also a real-number plan.

Check yourself

Q3. With 61 testing hours, check that $(10.5, 14.75)$ satisfies the three constraints, and that its profit is 1305 thousand DA.

3 What does "linear" mean?

Allowed (linear)	**Not allowed (not linear)**
a number times a variable: $40x_1$, $-2x_3$, $0.5x_2$	a product of variables: x_1x_2
sums of such terms: $40x_1 + 60x_2$	powers and roots: x_1^2, $\sqrt{x_1}$
a constant added: $3x_1 + 2x_2 + 10$	division by a variable: $1/x_1$
	absolute values: $\lvert x_1 - x_2 \rvert$

Examples. $3x_1 - x_2 \leq 5$ is linear. $x_1x_2 \geq 4$ is not linear (product). $x_1 + 2 = 7x_2$ is linear: it is the same as $x_1 - 7x_2 = -2$. Section 10 shows tricks to transform some non-linear conditions into linear ones.

A word about constants. Strictly speaking, an expression with a constant, like $3x_1 + 2x_2 + 10$, is called *affine*. Constants are no problem in a linear program: in an objective, a constant does not change the best plan (it only adds 10 to the value); in a constraint, we move it to the right side.

Comparison signs

A linear program uses only the signs $\leq$, $=$ and $\geq$. Strict inequalities $<$ and $>$ are *not allowed*, even if the expression is linear: $x_1 < 5$ is a linear inequality, but it cannot appear in an LP.

Why? With a strict inequality the best plan may not exist. Example: maximise x_1 subject to $x_1 < 5$. The plan 4.9 is feasible, 4.99 is better, 4.999 is better again, and there is no best plan (5 itself is not allowed). Solvers also accept only $\leq$, $=$ and $\geq$: for example, the Gurobi solver does not support strict inequalities. In practice we write $x_1 \leq 5$, or $x_1 \leq 4.99$ if the value 5 is really forbidden.

4 A method in five steps

Method 1.1 — Write a linear program

1. **Read** the text twice. Put all the numbers in a table, with units.
2. **Variables**: what do we decide? Write the meaning of each variable, with unit and period.
3. **Objective**: what do we want? Write z, and say max or min.
4. **Constraints**: what limits us? One constraint for each limit, each requirement and each balance.
5. **Check**: sign constraints, units in each constraint, every number of the text used, everything linear.

From words to constraints.

In the text	Constraint
"at most b", "not more than b", "we have b"	$\cdots \leq b$
"at least b", "not less than b", "we need b"	$\cdots \geq b$
"exactly b", "all of it must be used"	$\cdots = b$
"x is at least twice y"	$x \geq 2y$, i.e. $x - 2y \geq 0$
"at least 30% of the total is x_1"	$x_1 \geq 0.3\,(x_1 + x_2 + x_3)$ (Rule 1.1)
"x and y differ by at most 5"	$x - y \leq 5$ and $y - x \leq 5$ (Rule 1.7)
"what comes in = what goes out"	a balance: $\cdots = \ldots$

Good habit: put all the variables on the left side and the numbers on the right side.

5 Example 1.2: mixing raw materials

The problem. A factory makes exactly 1000 kg of a special material for robot arms, by mixing three raw materials, called M1, M2 and M3. Each raw material contains two important ingredients, called A and B. The final material must contain between 18% and 20% of ingredient A, and at least 8% of ingredient B. Find the cheapest mixture.

Raw material	Ingredient A	Ingredient B	Cost (DA/kg)	Available
M1	10%	2%	80	no limit
M2	5%	30%	600	300 kg
M3	60%	0%	250	no limit

Step 2: variables. x_1, x_2, x_3 = kilograms of M1, M2 and M3 in the mixture.

Step 3: objective. The cost is $80\,x_1 + 600\,x_2 + 250\,x_3$ dinars, to minimise.

Step 4: constraints.

- Total mass: exactly 1000 kg: $x_1 + x_2 + x_3 = 1000$.
- Ingredient A. 10% of x_1 kilograms is $0.10\,x_1$ kilograms of ingredient A; in the same way $0.05\,x_2$ and $0.60\,x_3$. So the mixture contains $0.10x_1 + 0.05x_2 + 0.60x_3$ kilograms of ingredient A. It must be between 18% and 20% of 1000 kg, that is between 180 and 200 kg. "Between" gives *two* constraints:

$$0.10x_1 + 0.05x_2 + 0.60x_3 \geq 180, \qquad 0.10x_1 + 0.05x_2 + 0.60x_3 \leq 200.$$

- Ingredient B: at least 8% of 1000 kg, that is 80 kg: $0.02x_1 + 0.30x_2 \geq 80$.
- Available M2: $x_2 \leq 300$. Signs: $x_1, x_2, x_3 \geq 0$.

The model.

$$\begin{aligned} \min \quad & z = 80x_1 + 600x_2 + 250x_3 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 1000 \\ & 0.10x_1 + 0.05x_2 + 0.60x_3 \geq 180 \\ & 0.10x_1 + 0.05x_2 + 0.60x_3 \leq 200 \\ & 0.02x_1 + 0.30x_2 \geq 80 \\ & x_2 \leq 300, \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

Here the masses can really be any real numbers, so this LP is the real problem (it is not a relaxation).

When the total is not fixed. Suppose now that the factory also decides how much material to make. Then "at least 18% of ingredient A" gives a quotient, which is not linear:

$$\frac{0.10x_1 + 0.05x_2 + 0.60x_3}{x_1 + x_2 + x_3} \geq 0.18.$$

Rule 1.1 — Percentages of a total

If the total is positive, multiply both sides by the total (an inequality does not change direction when we multiply by a positive number), then bring all terms to the left.

Here: $0.10x_1 + 0.05x_2 + 0.60x_3 \geq 0.18x_1 + 0.18x_2 + 0.18x_3$, and since $0.10 - 0.18 = -0.08$, $0.05 - 0.18 = -0.13$ and $0.60 - 0.18 = 0.42$:

$$-0.08x_1 - 0.13x_2 + 0.42x_3 \geq 0 \qquad \text{(linear)}.$$

Check with numbers. Take $x_1 = 500$, $x_2 = 200$, $x_3 = 300$ (total 1000 kg). Ingredient A: $50 + 10 + 180 = 240$ kg, that is 24%, at least 18%. The linear form gives $-40 - 26 + 126 = 60 \geq 0$. Both forms agree.

6 Example 1.3: transport

The problem. Two warehouses W1 and W2 store spare batteries: 300 in W1 and 400 in W2. Three sites S1, S2, S3, where robots work, need 200, 250 and 200 batteries. The cost of sending one battery is:

Cost (DA per battery)	S1	S2	S3	Stock
W1	800	600	1000	300
W2	900	1200	700	400
Need	200	250	200	

Find the cheapest transport plan.

Variables. We decide how many batteries travel on each route. There are $2 \times 3 = 6$ routes: x_{ij} = number of batteries sent from warehouse Wi to site Sj ($i = 1, 2$ and $j = 1, 2, 3$). For example, x_{12} is the number sent from W1 to S2.

The model.

$$
\begin{aligned}
\min \quad & z = 800x_{11} + 600x_{12} + 1000x_{13} + 900x_{21} + 1200x_{22} + 700x_{23} && \\
\text{s.t.} \quad & x_{11} + x_{12} + x_{13} \le 300 && \text{(what leaves W1)} \\
& x_{21} + x_{22} + x_{23} \le 400 && \text{(what leaves W2)} \\
& x_{11} + x_{21} \ge 200 && \text{(what S1 receives)} \\
& x_{12} + x_{22} \ge 250 && \text{(what S2 receives)} \\
& x_{13} + x_{23} \ge 200 && \text{(what S3 receives)} \\
& x_{ij} \ge 0 \text{ for all } i, j &&
\end{aligned}
$$

The sum symbol. $\sum_{j=1}^{3} x_{1j}$ means $x_{11} + x_{12} + x_{13}$. With this symbol the model can be written for any number of warehouses and sites: $\min \sum_i \sum_j c_{ij} x_{ij}$ subject to $\sum_j x_{ij} \le s_i$ (stock of warehouse i), $\sum_i x_{ij} \ge d_j$ (need of site j) and $x_{ij} \ge 0$.

A simple check. The total stock is $300 + 400 = 700$ and the total need is $200 + 250 + 200 = 650$. Since $650 \le 700$, there are enough batteries, and a feasible plan exists. For example: W1 sends 200 to S1 and 100 to S2, and W2 sends 150 to S2 and 200 to S3. W1 sends 300 (its whole stock), W2 sends 350, and each site receives exactly its need. The cost is $200 \times 800 + 100 \times 600 + 150 \times 1200 + 200 \times 700 = 540\,000$ DA.

If the sites needed more than 700 batteries in total, no plan could satisfy all the needs: the problem would have no feasible solution (we say it is *infeasible*).

Check yourself

Q4. Give another feasible plan for Example 1.3 and compute its cost. Is it cheaper than 540 000 DA?

7 Example 1.4: assignment

The problem. Three robots R1, R2, R3 must do three tasks T1, T2, T3. Each robot does exactly one task, and each task is done by exactly one robot. The times (in minutes) are:

Time (minutes)	T1	T2	T3
R1	9	2	7
R2	6	4	3
R3	5	8	1

Find the assignment with the smallest total time.

How many assignments? For T1 we can choose 3 robots; then 2 robots remain for T2; then 1 for T3. So there are $3 \times 2 \times 1 = 6$ assignments. With n robots and n tasks there are $n! = n \times (n-1) \times \cdots \times 1$ assignments: $10! = 3\,628\,800$ and $20! \approx 2.4 \times 10^{18}$. We cannot try them all.

Variables. $x_{ij} = 1$ if robot Ri does task Tj, and $x_{ij} = 0$ otherwise (9 variables).

The model.

$$\min\ z = 9x_{11} + 2x_{12} + 7x_{13} + 6x_{21} + 4x_{22} + 3x_{23} + 5x_{31} + 8x_{32} + 1x_{33}$$

subject to

- each robot does one task: $x_{11} + x_{12} + x_{13} = 1,\ x_{21} + x_{22} + x_{23} = 1,\ x_{31} + x_{32} + x_{33} = 1$;
- each task is done by one robot: $x_{11} + x_{21} + x_{31} = 1,\ x_{12} + x_{22} + x_{32} = 1,\ x_{13} + x_{23} + x_{33} = 1$;
- $x_{ij} \geq 0$.

Only the chosen pairs (with $x_{ij} = 1$) count in z. The condition "$x_{ij} = 0$ or 1" is not linear, so we write only $x_{ij} \geq 0$; the equalities then give $x_{ij} \leq 1$. We accept the following fact without proof: *for this problem, the linear program always has an optimal solution with all values equal to 0 or 1*, so it gives a real assignment.

8 Writing a linear program with matrices

For the drone workshop, put the variables, the profits and the data in columns and in a matrix:

$$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad c = \begin{pmatrix} 40 \\ 60 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \\ 0 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 40 \\ 60 \\ 18 \end{pmatrix}.$$

The model is then: $\max\ c^\top x$ subject to $Ax \leq b,\ x \geq 0$.

How to read it.

- $c^\top x = 40x_1 + 60x_2$: line times column.
- Ax is computed line by line: each line of A times the column x:

$$Ax = \begin{pmatrix} 1 \cdot x_1 + 2 \cdot x_2 \\ 3 \cdot x_1 + 2 \cdot x_2 \\ 0 \cdot x_1 + 1 \cdot x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ 3x_1 + 2x_2 \\ x_2 \end{pmatrix}.$$

- $Ax \leq b$ means: each line of Ax is at most the same line of b. So one line of A is one constraint.
- One column of A tells what one unit of a variable uses: column 1 says that one drone A uses 1 hour of assembly, 3 hours of testing and 0 camera.
- $x \geq 0$ means $x_1 \geq 0$ and $x_2 \geq 0$.

9 Different forms of the same linear program

Definition 1.3 — Canonical form

$\max\ c^\top x$ subject to $Ax \leq b,\ x \geq 0$. All constraints are "$\leq$", all variables are ≥ 0, and we maximise.

Definition 1.4 — Standard form

$\max\ c^\top x$ subject to $Ax = b,\ x \geq 0$. All constraints are equalities, all variables are ≥ 0, and we maximise.

The drone workshop is already in canonical form. The standard form is the form used by the simplex method, which we will study later. Every linear program can be written in both forms with the following rules.

Rule 1.2 — Minimum into maximum

min z is the same as max $(-z)$: the best plan is the same, and the optimal value only changes sign. Example: min $3x_1 + x_2$ becomes max $-3x_1 - x_2$. If the minimum of $3x_1 + x_2$ is 5, the maximum of $-3x_1 - x_2$ is -5, at the same plan.

Rule 1.3 — Changing the direction of an inequality

Multiply both sides by -1: "$\geq$" becomes "$\leq$" (multiplying by a negative number reverses an inequality). Example: $x_1 + x_2 \geq 2$ becomes $-x_1 - x_2 \leq -2$.

Rule 1.4 — An equality is two inequalities

$a = b$ is the same as "$a \leq b$ and $a \geq b$". Example: $x_1 - x_2 = 1$ becomes $x_1 - x_2 \leq 1$ and $x_1 - x_2 \geq 1$ (and then $-x_1 + x_2 \leq -1$ by Rule 1.3).

Rule 1.5 — A free variable

A variable that may be negative is called *free*. Write it as the difference of two non-negative variables: $x_j = x_j^+ - x_j^-$ with $x_j^+ \geq 0$, $x_j^- \geq 0$, and replace x_j everywhere. Example: $3 = 3 - 0$ (so $x_j^+ = 3$, $x_j^- = 0$), and $-2 = 0 - 2$ (so $x_j^+ = 0$, $x_j^- = 2$).

Rule 1.6 — Slack and surplus variables

A constraint "$\leq$" becomes an equality if we *add* a slack variable $s \geq 0$ (the unused quantity). A constraint "$\geq$" becomes an equality if we *subtract* a surplus variable $e \geq 0$ (the quantity above the requirement). Examples: $x_1 + 2x_2 \leq 40$ becomes $x_1 + 2x_2 + s_1 = 40$, $s_1 \geq 0$; $x_1 + x_2 \geq 2$ becomes $x_1 + x_2 - e_1 = 2$, $e_1 \geq 0$.

Worked example (a): the drone workshop in standard form. We add one slack variable per constraint: s_1 = unused assembly hours, s_2 = unused testing hours, s_3 = unused cameras.

$$\begin{aligned} &\max\ 40x_1 + 60x_2 \\ &\text{s.t. } x_1 + 2x_2 + s_1 = 40, \quad 3x_1 + 2x_2 + s_2 = 60, \quad x_2 + s_3 = 18, \\ &\qquad x_1, x_2, s_1, s_2, s_3 \geq 0. \end{aligned}$$

For the plan $(10, 10)$: $s_1 = 40 - 30 = 10$ unused assembly hours, $s_2 = 60 - 50 = 10$ unused testing hours, $s_3 = 18 - 10 = 8$ unused cameras.

Worked example (b): from a general LP to the canonical form.

$$\min\ 3x_1 + 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2, \quad x_1 - 2x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \text{ free.}$$

1. Rule 1.2: min $3x_1 + 2x_2$ becomes max $-3x_1 - 2x_2$.
2. Rule 1.5: $x_2 = x_2^+ - x_2^-$. The objective becomes $-3x_1 - 2x_2^+ + 2x_2^-$; the constraints become $x_1 + x_2^+ - x_2^- \geq 2$ and $x_1 - 2x_2^+ + 2x_2^- \leq 4$.
3. Rule 1.3 on the first constraint: $-x_1 - x_2^+ + x_2^- \leq -2$.

Canonical form:

$$\begin{aligned} &\max\ -3x_1 - 2x_2^+ + 2x_2^- \\ &\text{s.t. } -x_1 - x_2^+ + x_2^- \leq -2, \quad x_1 - 2x_2^+ + 2x_2^- \leq 4, \quad x_1, x_2^+, x_2^- \geq 0. \end{aligned}$$

(The standard form of this example is an exercise of TD 1.)

This problem has a best plan: $(x_1, x_2) = (0, 2)$ with $z = 4$. Indeed, the first constraint gives $x_2 \geq 2 - x_1$, so $3x_1 + 2x_2 \geq 3x_1 + 4 - 2x_1 = x_1 + 4 \geq 4$, and $(0, 2)$ reaches 4 (check: $0 + 2 \geq 2$ and $0 - 4 \leq 4$).

Warning: changing the form does not tell whether a best plan exists. Keep the same constraints, but take the objective min $3x_1 - x_2$. For $x_1 = 0$ and $x_2 = t$ with $t \geq 2$, both constraints hold ($t \geq 2$ and $-2t \leq 4$), and the objective is $-t$, which goes to $-\infty$ when t grows. This LP has no best plan: it is *unbounded* (Lecture 2, Case 4). The conversion rules work in the same way, but there is no optimal plan to find.

10 Useful tricks, and what is impossible

Rule 1.7 — Absolute value "at most"

$|a| \leq b$ is the same as "$a \leq b$ and $-a \leq b$". For example, $|x_1 - x_2| \leq 5$ becomes $x_1 - x_2 \leq 5$ and $x_2 - x_1 \leq 5$.

Why? The absolute value of a number is the larger of the number and its opposite: $|6| = 6$, $|-6| = 6$. The larger of two numbers is at most 5 exactly when both are at most 5. *Check.* $x = (8, 2)$: $|8 - 2| = 6 > 5$, and indeed $x_1 - x_2 = 6 > 5$, so the linear form also says "no". $x = (4, 7)$: $|4 - 7| = 3 \leq 5$, and $4 - 7 = -3 \leq 5$, $7 - 4 = 3 \leq 5$: both forms say "yes". *Warning.* $|x_1 - x_2| \geq 5$ ("at least") can *not* be transformed like this.

Rule 1.8 — Maximise the smallest value

To maximise $\min(x_1, x_2)$ (for example, to be fair between two robots), add a new variable t: maximise t subject to $t \leq x_1$, $t \leq x_2$ (and the other constraints).

Why? The constraints say that t is at most both numbers, so t is at most the smaller one. Since we maximise t, the best plan pushes t up until it is equal to the smaller one. In the same way, to *minimise* the largest value $\max(x_1, x_2)$, minimise t subject to $t \geq x_1$ and $t \geq x_2$.

What is impossible: products. A condition like $x_1 x_2 \leq 1$ (with $x_1, x_2 \geq 0$) can never be written with linear constraints. Here is a simple reason.

- *Linear constraints keep middle points.* If two points P and Q satisfy a linear constraint, for example $a_1 x_1 + a_2 x_2 \leq b$, then the middle point $M = \frac{P+Q}{2}$ also satisfies it: the value of $a_1 x_1 + a_2 x_2$ at M is the average of its values at P and Q, and the average of two numbers $\leq b$ is $\leq b$.
- *The product condition does not.* The points $P = (2, 0.5)$ and $Q = (0.5, 2)$ satisfy $x_1 x_2 = 1 \leq 1$. Their middle point is $M = (1.25, 1.25)$, and $1.25 \times 1.25 = 1.5625 > 1$.

So no list of linear constraints can describe exactly the condition $x_1 x_2 \leq 1$.

11 The four hypotheses of linear programming

A linear program is a simplified picture of reality. It assumes:

- **Proportionality**: two drones bring twice the profit of one drone (no discount for large quantities).

- **Additivity**: the total is the sum of the parts (the profit of A does not depend on the number of B).
- **Divisibility**: the variables can take any real value (10.5 drones is allowed by the model: this is the LP relaxation of Section 2.5).
- **Certainty**: all the numbers of the model are known exactly.

When a hypothesis is not true we use other tools: *integer programming* when we need whole numbers, and *sensitivity analysis* (Chapter 3) when the numbers are uncertain.

12 Summary

- A linear program has variables, a linear objective (max or min), linear constraints and sign constraints (Definition 1.1).
- Feasible solution, feasible set, optimal solution, optimal value (Definition 1.2).
- To write an LP, follow Method 1.1: read, variables, objective, constraints, check.
- Four classical models: production (drones), mixing (raw materials), transport (batteries), assignment (robots and tasks).
- Whole-number problems: the LP is a *relaxation*; its answer is whole only by luck (Section 2.7).
- Matrix form $\max c^\top x$, $Ax \leq b$, $x \geq 0$; canonical form (Definition 1.3) and standard form (Definition 1.4); Rules 1.2 to 1.6 to change the form.
- Tricks: percentages (Rule 1.1), absolute values (Rule 1.7), maximising the smallest value (Rule 1.8). Products of variables are impossible.

Reading. F. S. Hillier and G. J. Lieberman, *Introduction to Operations Research*, Chapter 3; H. A. Taha, *Operations Research: An Introduction*, Chapter 2.

Answers to "Check yourself"

Q1. $(12, 12)$: assembly $12 + 24 = 36 \leq 40$, testing $36 + 24 = 60 \leq 60$, cameras $12 \leq 18$: feasible. Profit $40 \times 12 + 60 \times 12 = 1200$ thousand DA.

Q2. $(20, 2)$: assembly $20 + 4 = 24 \leq 40$, but testing $60 + 4 = 64 > 60$: not feasible.

Q3. Assembly $10.5 + 29.5 = 40 \leq 40$, testing $31.5 + 29.5 = 61 \leq 61$, cameras $14.75 \leq 18$. Profit $40 \times 10.5 + 60 \times 14.75 = 420 + 885 = 1305$ thousand DA.

Q4. Many answers. For example: W1 sends 50 to S1 and 250 to S2; W2 sends 150 to S1 and 200 to S3. Check: W1 sends 300, W2 sends 350; S1 receives $50 + 150 = 200$, S2 receives 250, S3 receives 200. Cost: $50 \times 800 + 250 \times 600 + 150 \times 900 + 200 \times 700 = 40\,000 + 150\,000 + 135\,000 + 140\,000 = 465\,000$ DA, cheaper than 540 000 DA. (In fact, this is the cheapest plan.)